EXP NO:1 STUDY OF VARIDUS NETWORK COMMANDS
23/7/26 USED IN LINUX AND WINDOWS

AIM:

Study of various Network commands used in Linux and Windows.

Basic Networking Commands:

1) arp-a: ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

2) hostname: This is the simplest of all TCP/IP commands. it simply displays the name of your computer.

1st output:

Interface: 172.16.10.127 --- 0xe

Internet Address	Physical address	Type
172.16.8.1	7c-5a-1c-cf-be-45	dynamic
172.16.8.5	c0-35-32-70-f4-1b	dynamic
172.16.8.9	d4-3d-7e-ad-40-6a	dynamic

2nd O/P: DESKTOP-HIMFLV4

3) ipconfig/all: This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of ethernet adapter in your system.

O/P: WINDOWS IP configuration

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Link-Local IPV6 Address . . . . . : fe80::ea1c:acfd:c683:58c1%14
IPV4 address . . . . . : 172.16.10.127
subnet mask 255.255.252.0
Default gateway 172.16.8.1

4) nbtstat -a : This command helps solve problems with NETBIOS name resolution. (Nbt stands for NetBIOS over TCP/IP)

O/P:

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [Interval]]

-a (adapter status) Lists the remote machine's name table given its name

-A (adapter status) Lists the remote's machine's name table given it's IP address.

-c (cache) Lists NBT's cache of remote names and their IP address.

5) netstat : (network statistics) netstat displays variety of statistics about a computers active TCP/IP connections. it is a command line tool for monitoring connections both incoming and outgoing as well as viewing routing tables, interface statistics., eg: netstat -r

O/P:

Active Connections

Proto	Local Address	Foreign address	state
TCP	172.16.10.127:64151	4.213.25.240:https	ESTABLISHED
TCP	172.16.10.127:64402	sf-in-f188:5228	ESTABLISHED
TCP	172.16.10.127.64453	a23-44-10-58:https	CLOSE_WAIT

6) nslookup : (name server lookup) is a tool used to perform DNS lookups in Linux. it is used to display DNS details, such as IP address of a particular computer, the mx records for a domain or NS servers, two modes: interactive and non-interactive.

O/P:

Default Server : Unknown

Address : 172.16.8.1

7) pathping: Pathpig is unique to window's and is basically a combination of ping and Tracert commands. pathping traces the routes to destination address then launches a 25 test of each router along the way, gathering data loss along each hop.

O/P:

Pathping [-g host-list] [-h maximum_hops] [-i address] [-n] [-p period] [-q num_queries] [-w timeout] [-4] [-6] target_name

options:

-g host-list Loose source route along host-list.

-h maximum_hops Maximum no. of hops to search for target.

8) ping: (Packet INternet Groper) command is the best way to test connectivity between two nodes.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name. (ping www.facebook.com)

O/P:

ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS] [-r count] [-s count] [[-j host-list] [-k host-list]] [-w timeout] [-R] [-S srcaddr] [-P] [-4] [-6] target_name.

9) Route: route command is used to show/manipulate the IP routing table. Primarily to setup or networks via interface.

O/P:

ROUTE [-f] [-p] [-4|-6] command [destination] [MASK netmask] [gateway] [if interface]

-f clears the routing tables of all gateway. if this is used in conjunction with one of the commands, the tables are cleared prior to running the command.

10) ip address show: Displays all network interfaces and their assigned ip address.

O/P: enp0s3 UP

11) ip address add
192.168.1.254/24 dev enp0s03

Assigns the IP address 192.168.1.254 to interface enp0s03.

O/P:

12) ip address del 192.168.1.254/24 dev enp0s03

Removes the specified ip address from the interface

O/P:

13) ip link set eth0 down

Disables (deactives) the network interface eth0.

O/P:

14) ip link set eth0 down

Disables (deactives) the network interface eth0.

O/P:

15) ip link set eth0 promisc on

Enables promiscous mode so the interface captures all networks packets.

o/p:

2) ifconfig: The ifconfig command was a staple in many sysadamin's tool belt for configuring and troubleshooting networks. it has been replaced by the ip command discussed above.

o/p:

```
eth0: flags = 4163 <UP, BRODCAST, RUNNING,
      MULTICAST > mtu 1500
      inet 192.168.1.100 netmask 255.255.0
      broadcast 192.168.1.255
      inet6 fe80::a00:27ff:fe4e prefixlen 64
      scopeid 0x20 <link>
      ether 08:00:27:4e:66:a1 (1000 (ethernet)
```

3) mtr: MTR is a progress with command line interface that serves as a network diagonistic and Troubleshooting tool.

o/p:

My traceroute [v0.87]

Host	Packets						
1.1::1	Loss%	snt	last	Avg	Best	st dev	
	0.0%	145	0.2	0.1	0.1	0.6	0.0

4) tcpdump:

The tcpdump command is designed for capturing and displaying packets.

o/p:

tcpdump : you don't have persimission to capture on that device.

5) ping:

ping is a tool that verifies IP_level connectivity to another TCP/IP computer by sending interface control message.

o/p:

PING google.com (142.200.182.78) 56(84) bytes of data 64 bytes from lemma-arc-in = 1 tti = 112 time = 988 ms.

Result:

Thus the basic networking and linux commands are learned and o/p is got verified.